

Expressway Architectural Update

cisco Live !

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Webex App

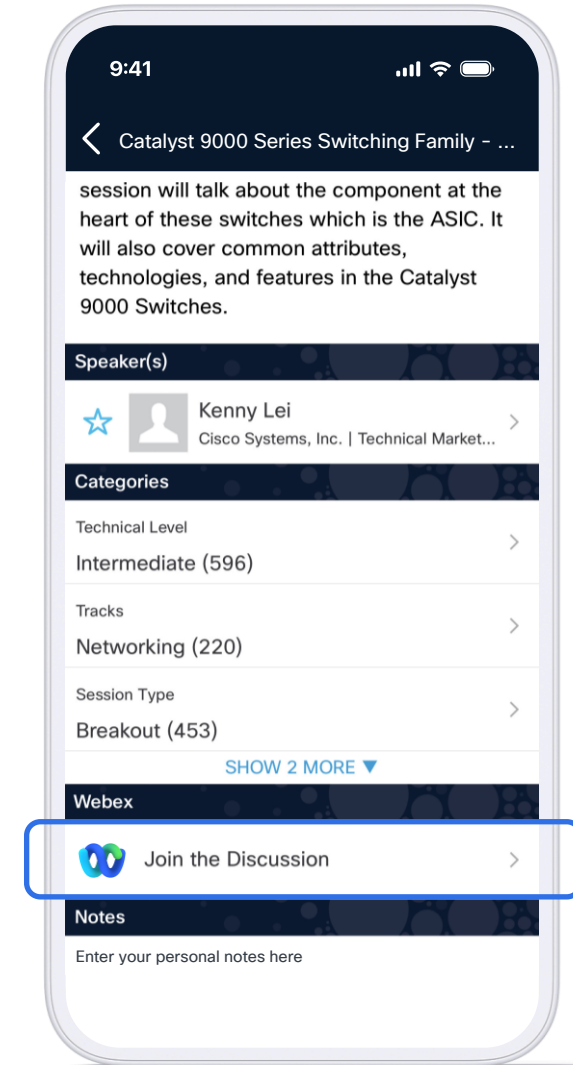
Questions?

Use Webex App to chat with the speaker after the session

How

- 1 Find this session in the Cisco Events App
- 2 Click “Join the Discussion”
- 3 Install the Webex App or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

Webex spaces will be moderated by the speaker until February 27, 2026.



Objectives

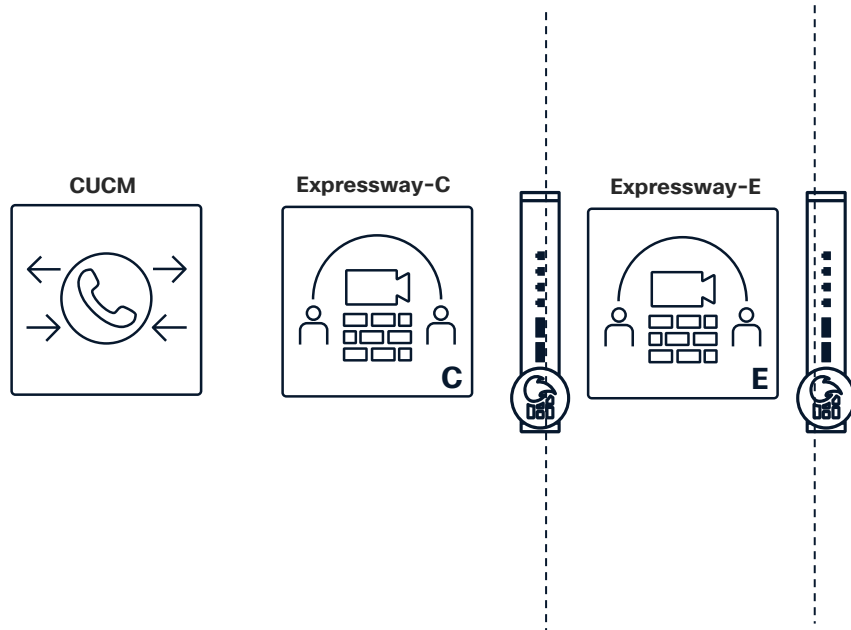
- Understand endpoint failover and redundancy
- Understand the implications of ECU ClientAuth Sunset
- Understand platforms and cluster redundancy options

Agenda

- 01 Introduction**
- 02 Platforms and Scalability**
- 03 MRA Architectures**
- 04 Cloud And On-Prem Devices**
- 05 Client Auth ECU Sunset**
- 06 Security**
- 07 CTI Over MRA**
- 08 VDI Architectures**

Introduction

Expressway Services



B2B Calling

Mobile and
Remote Access

CMS WebRTC

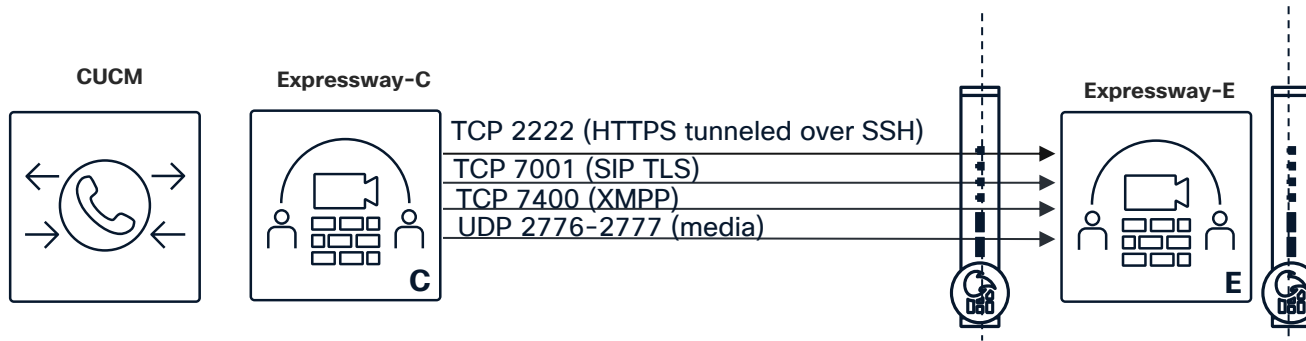
Interworking

XMPP
Federation

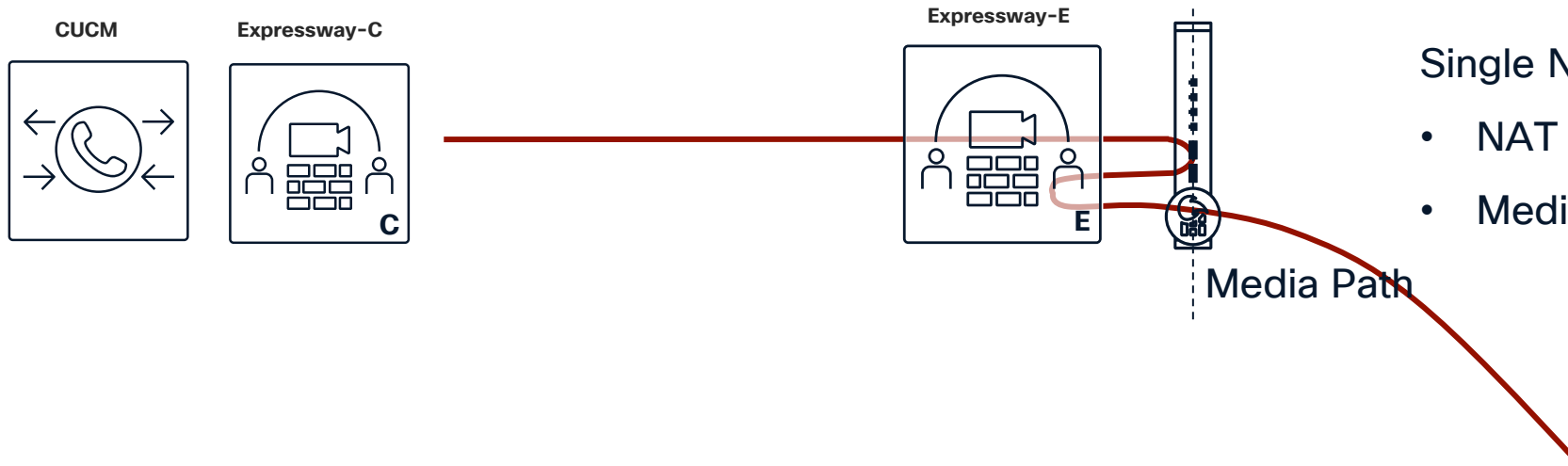
Call Control

Webex Edge

Expressway Network Deployment Models



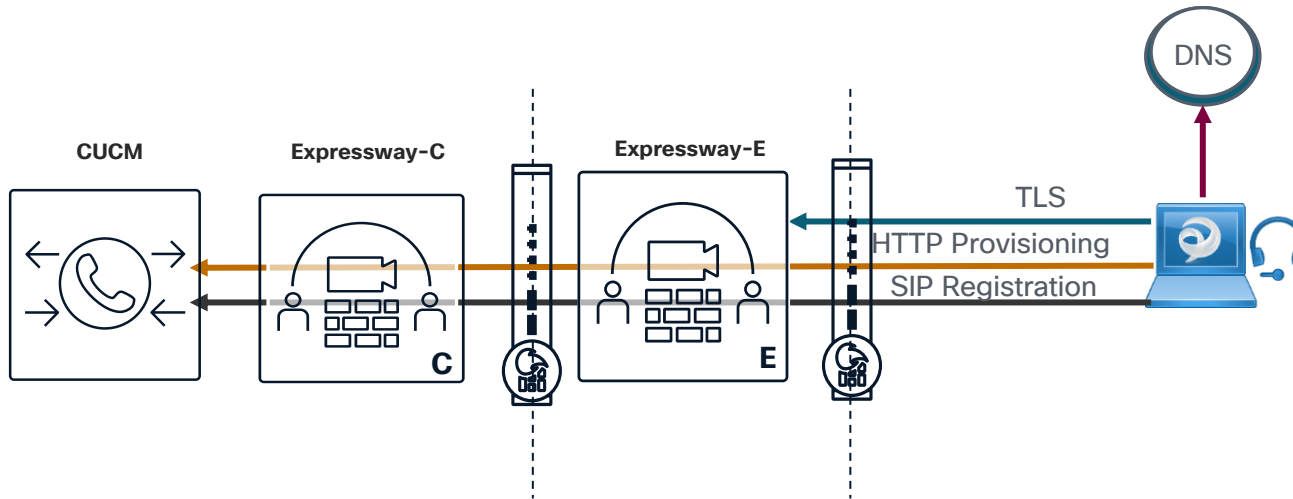
Dual NIC - Preferred



Single NIC - Supported But Not Recommended

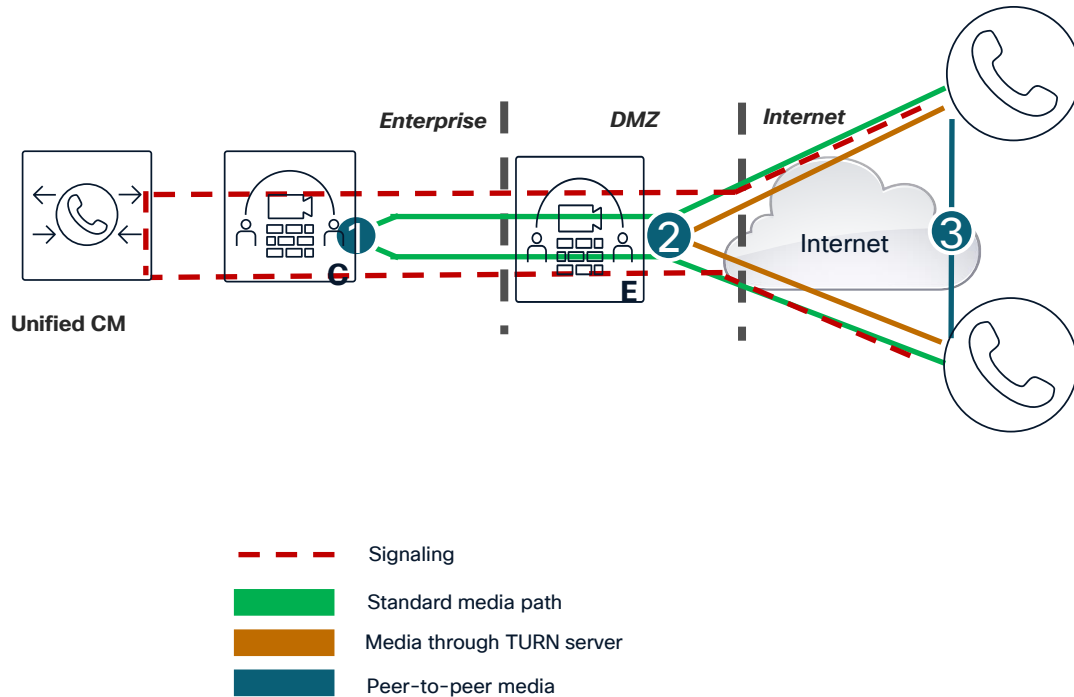
- NAT Reflection Required
- Media Streams x 3 in the firewall

Expressway Deployment For MRA



- 1 Service Discovery through DNS SRV
- 2 Connection setup via TLS on port 8443. Endpoints and sw clients check Expressway identity and domain through certs
- 3 Provisioning done via HTTP on top of TLS connection (HTTPS), and HTTPS over SSH between Expressway-E and Expressway-C
- 4 Registration through SIP/XMPP

MRA Signaling and Media Flows



SIP Signaling

- 1 Standard media path** – This is the traditional media path for MRA endpoints (without ICE enablement)
- 2 Media through the TURN server** – In this case the media is bridged between the MRA endpoints and a TURN server(s) over their direct Internet access.
- 3 Peer-to-peer media** – This is a media path for ICE-enabled endpoints where media is able to flow directly between the endpoints.

Platforms and Scalability

Virtual Machine Requirements

	vCPU	Physical CPU Base Frequency	vRAM	vDisk (GB)	vNIC
Small	2	1.80	4	132	1 Gb
Medium	2	2.40	6	132	1 Gb
Large	8	3.20	8	132	1 Gb

- No oversubscription of CPU. 1:1 allocation of vCPU to physical cores
- No oversubscription of RAM. 1:1 allocation of vRAM to physical memory
- No oversubscription of NIC

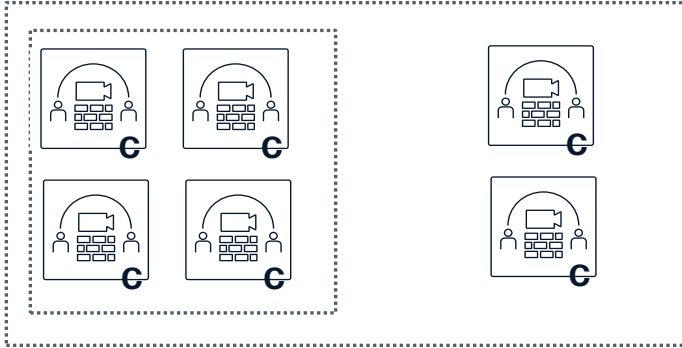
Expressway Capacity

X15 with Fast Path

Platform	Calls (Video or Audio-only)	RMS Licenses	MRA Registrations (proxied)
Large VM	500 video or 1000 audio	500	4000
Medium VM	150 video or 300 audio	150	3000
Small VM	20 video or 40 audio	75	200

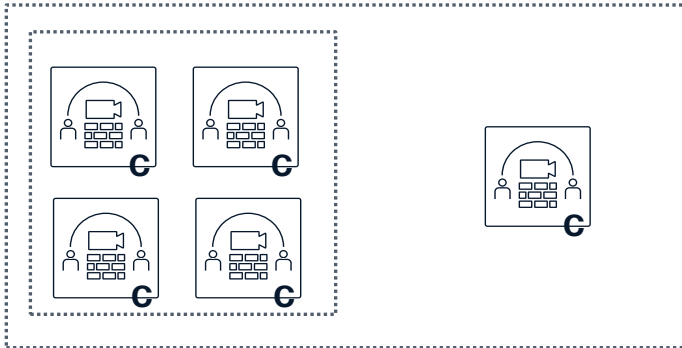
Clustering Guidelines For Expressway-C and Expressway-E

Expressway-C and Expressway-E Redundancy Models



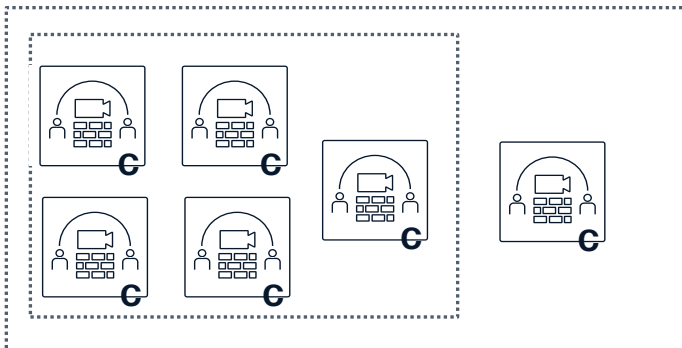
- 4+2 Redundancy Model: capacity of a single server x 4
- All Releases

Equal number of peers in Expressway-C Expressway-E clusters



- 4+1 Redundancy Model: capacity of a single server x 4
- From Release X14.2

Same OVA size throughout cluster



- 5+1 Redundancy Model: capacity of a single server x 5
- From release X14.2

Same software version throughout cluster

Clustered Expressway Capacity

- Let X:Y be the capacity model, and X the number of active servers
- Let S be the quantity of video calls, licenses or MRA registrations per server
- Let C be the quantity of video calls, licenses or MRA registrations per cluster
- Then $C = X \times S$
- Examples

Platform	Clustering Model	Calls (Video or Audio-only)	RMS Licenses	MRA Registrations (proxied)
Large VM	4:2/4:1	2000 video or 4000 audio	2000	16000
Large VM	5:1	2500 video or 5000	2500	20000
Medium VM	4:2/4:1	600 video or 1200 audio	600	12000
Medium VM	5:1	750 video or 1500 audio	750	15000

Installed Base Hardware Refresh/Add-on BE6K/BE7K & CE1400V Appliances

Business Edition
6000 / 7000
Appliance
Models



ESXi

BE6000M M6/M5
BE6000H M5
(M4/older eoSupport)



BE6000 (M7)
BE6K-M7-K9



*NFVIS or
ESXi*



ESXi

BE7000M M6/M5
BE7000H M6/M5
(M4/older eoSupport)



BE7000M (M7)
BE7M-M7-K9



*NFVIS or
ESXi*



*NFVIS or
ESXi*



bare-metal

CE1300 (M6)
CE1200 (M5)
(CE1100/500 eoSupport)



CE1400V (M7)
CE1400V-M7-K9



*NFVIS or
ESXi*

3 Large VM or
6 Medium VM

CE1400V (1)

Security Best Practices for Virtualization in the DMZ

- **Reduce Virtualization Risk**

- Deploy **one VM per host** to eliminate lateral movement.
- Treat ESXi as a **single-purpose appliance**.
- Manage the host as a standalone system (no vCenter, no vMotion, no clustering).
- Enforce strict mapping: **Physical NIC : vSwitch : Port Group : vNIC = 1 : 1 : 1 : 1**

- **Management Access Only From Inside**

- ESXi management (HTTPS, SSH, APIs) allowed **only** from internal trusted management networks.
- Firewall rule: **internal → ESXi mgmt**, deny all other paths.
- ESXi patching and upgrades allowed **only** from internal networks.
- Use a **Dedicated Management Interface (DMI) on Expressway** for appliance administration.

CE1400V (2)

Security Best Practices for Virtualization in the DMZ

- **Logging and Monitoring**

- Forward ESXi logs **only to an internal syslog server** (or SIEM if available).
- Monitor all management access attempts (auth failures, shell usage, API calls).
- Track state changes (services, network configuration, host operations).

MRA Architectures

Provisioning

- Provisioning happens via HTTP and is an endpoint-driven process
- Endpoint asks for *get_edge_config* request, and receives *getEdgeConfigResponse* from Expressway-C.
- *getEdgeConfigResponse* includes the servers that are available to handle that registration. Least loaded first.

```
<service>
|   <name>tftpServer</name>
|   <address>us_ea_ccm001.cloudentpa.com</address>
|   <address>us_we_ccm002.cloudentpa.com</address>
|
|-----> TFTP Servers
</service>
</serviceConfig>
<edgeConfig>
  <sipEdgeServer>
    <server>
      <address>us_ea_expe001.cloudentpa.com</address>
      <tlsPort>5061</tlsPort>
    </server>
    <server>
      <address>us_ea_expe003.cloudentpa.com</address>
      <tlsPort>5061</tlsPort>
    </server>
    <server>
      <address>us_ea_expe005.cloudentpa.com</address>
      <tlsPort>5061</tlsPort>
    </server>
  </sipEdgeServer>
  <sipRequest>
    <route>&lt;sip:10.99.101.201:5061;transport=tls;zone-id=1;directed;lr&gt;</route>
    <route>&lt;sip:10.99.101.203:5061;transport=tls;zone-id=1;directed;lr&gt;</route>
    <route>&lt;sip:10.99.101.202:5061;transport=tls;zone-id=1;directed;lr&gt;</route>
  </sipRequest>
  -----> Expressway-E (SIP)
  -----> Expressway-C
```

SIP Registration Through Expressway-E

- Endpoints and clients define the registration path based on the *getEdgeConfigResponse*
- Route Header includes the full registration path, specified hop-by-hop
- Route Header takes the precedence over the request URI

```
1 SIPMSG:
2 |REGISTER sip:us_we_ccm002.cloudentpa.com SIP/2.0
3 Via: SIP/2.0/TLS 192.168.1.224:57852;branch=z9hG4bK0000578b
4 Call-ID: f82fa8e4-a6060002-00004d85-00002981@192.168.1.224
5 CSeq: 103 REGISTER
6 Contact: <sip:a4d71c16-61af-06b2-3ddb-9b44c50384b6@192.168.1.224:57852;transport=tls>;+sip.instance="<urn:uuid:00000000-0000-0000-0000-f82fa8e4a606>";+u.sip!devicename.ccm.cisco.
7 From: <sip:+19725555105@us_we_ccm002.cloudentpa.com>;tag=f82fa8e4a606000300004f85-00000f9d
8 To: <sip:+19725555105@us_we_ccm002.cloudentpa.com>
9 Max-Forwards: 70
10 Route: <sip:us_ea_expe001.cloudentpa.com;transport=tls;lr>,<sip:10.99.101.201:5061;transport=tls;zone-id=1;directed;lr>,<sip:us_we_ccm002.cloudentpa.com:5091;transport=tls;lr>
11 User-Agent: Cisco-CSF
12 Expires: 0
13 Date: Mon, 04 May 2020 14:38:22 GMT
14 Supported: replaces,join,sdp-anat,norefersub,resource-priority,extended-refer,X-cisco-callinfo,X-cisco-serviceuri,X-cisco-escapecodes,X-cisco-service-control,X-cisco-srtp-fallback
15 Content-Length: 0
```

SIP Registration Through Expressway-C

- Expressway-E removes the previous Route Headers
- UCM Route Header only left on Expressway-C
- UCM receives the message and processes the registration

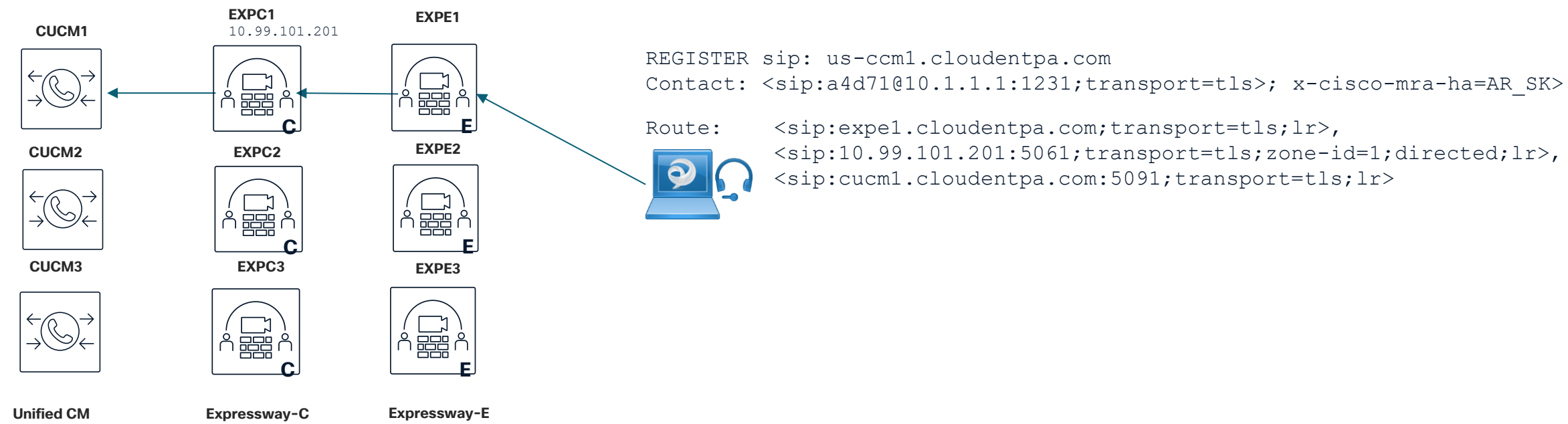
```
SIPMSG:
|REGISTER sip:us_we_ccm002.cloudentpa.com SIP/2.0
Via: SIP/2.0/TLS 203.0.113.241:7001;egress-zone=mratraversalserver;branch=z9hG4bK20
Via: SIP/2.0/TLS 192.168.1.224:57852;branch=z9hG4bK00002900;received=198.51.100.112
Call-ID: f82fa8e4-a6060002-00004d85-00002981@192.168.1.224
CSeq: 102 REGISTER
Contact: <sip:a4d71c16-61af-06b2-3ddb-9b44c50384b6@192.168.1.224:57852;transport=t
From: <sip:+19725555105@us_we_ccm002.cloudentpa.com>;tag=f82fa8e4a606000300004f85-0
To: <sip:+19725555105@us_we_ccm002.cloudentpa.com>
Max-Forwards: 15
Route: <sip:us_we_ccm002.cloudentpa.com:5091;transport=tls;lr>
Path: <sip:203.0.113.241:7001;transport=tls;lr>
Path: <sip:198.51.100.112:53782;transport=tls;apparent;ds;lr>
User-Agent: Cisco-CSF
Expires: 0
Date: Mon, 04 May 2020 14:38:21 GMT
Supported: replaces,join,sdp-anat,norefersub,resource-priority,extended-refer,X-cis
P-Asserted-Identity: <sip:+19725555105@us_we_ccm002.cloudentpa.com>
X-TAATag: 9ae26fd5-4a2b-4b7d-b78a-f395b0046918
Session-ID: 8afcebd800255000a00009b5fb2f0000;remote=00000000000000000000000000000000
Mime-Version: 1.0
Content-Type: multipart/mixed;boundary=uniqueBoundary
Content-Length: 1271
```

SIP Signaling Failover

- Failover mechanism based on
 - Adaptive Routing
 - STUN Keepalives
- For failover to occur properly, both servers and clients must support the above mechanisms

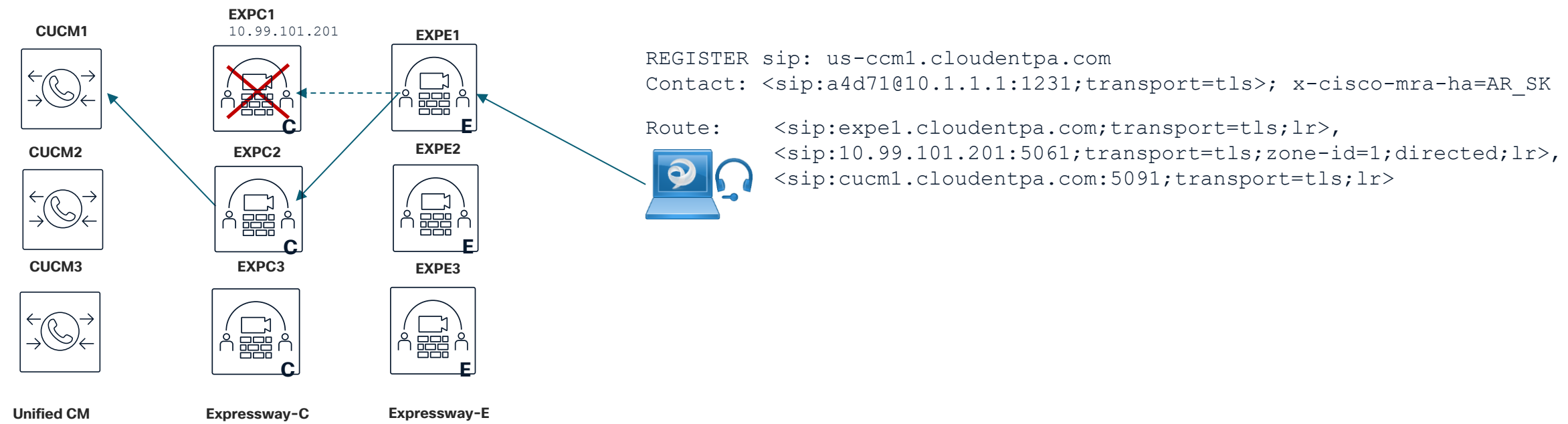
Adaptive Routing: Signaling Failover During Registration

SIP Registration



Adaptive Routing: Endpoint Failover During Registration

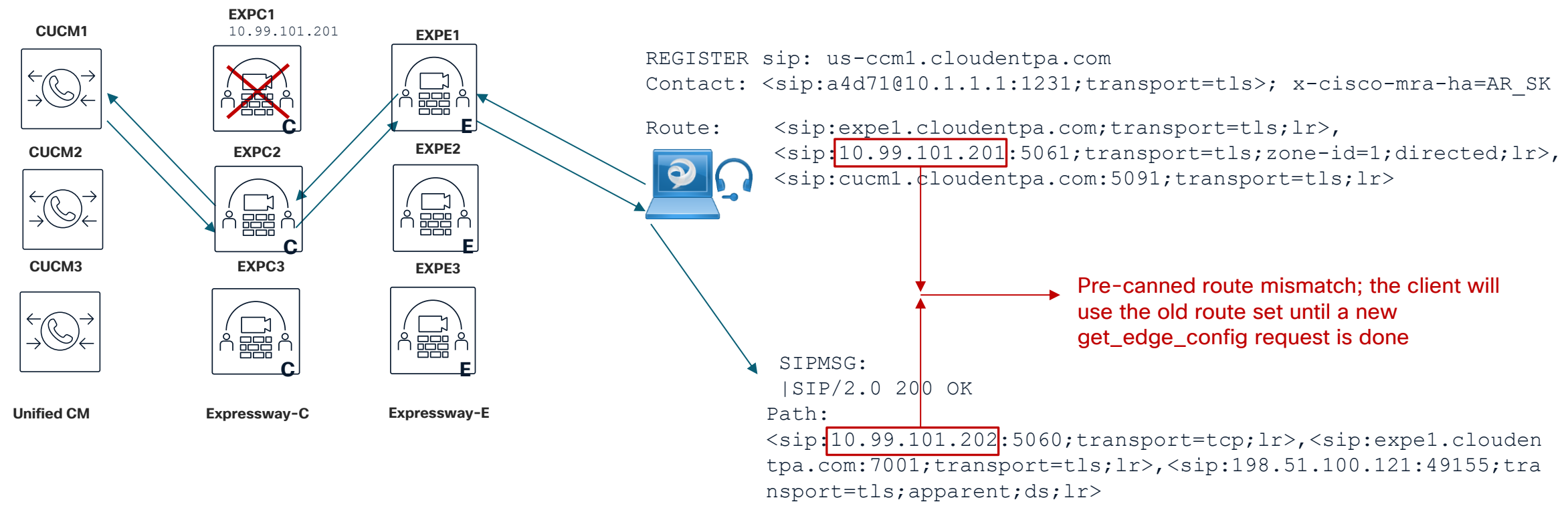
SIP Registration with Expressway-C Failure



Adaptive Routing: Endpoint Failover During Registration

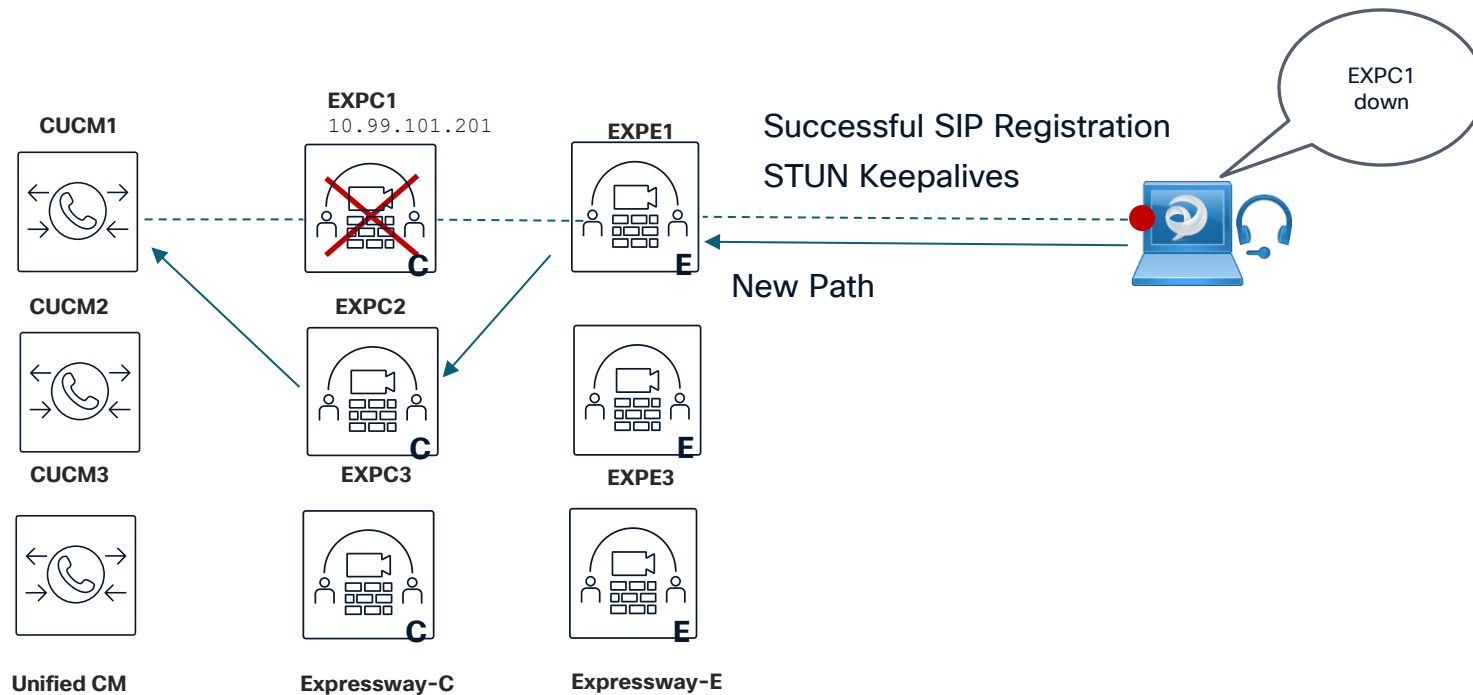
Endpoint pre-canned route update

- Path Header in 200 OK specifies the new route



STUN Keepalives: Signaling Failover After Registration

- HW and SW clients send STUN packets (Binding Requests) every 30 seconds
- If successful, Binding Responses will be received by the client
- If unsuccessful, a new STUN attribute will be sent by the server containing the failed node in the path

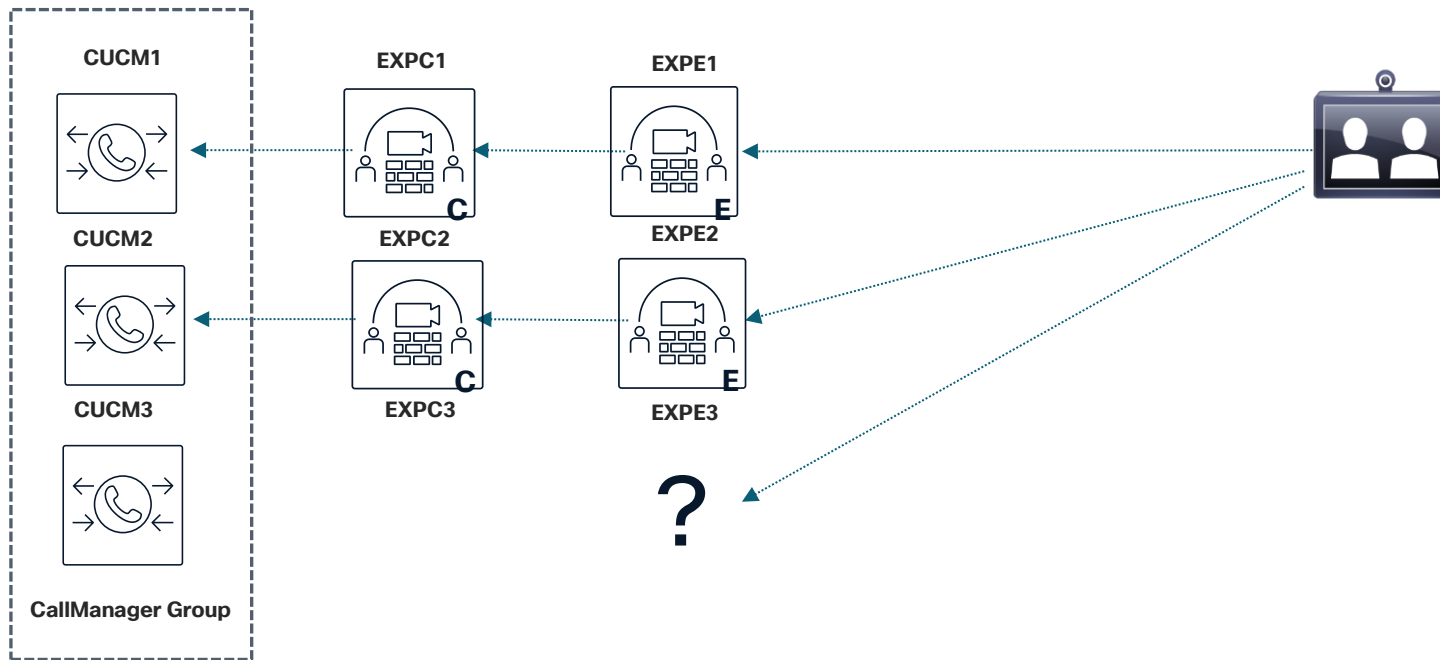


STUN Keepalives and Adaptive Routing Releases

- Cisco Jabber release 12.9 MR and above
- Webex app
- 78XX and 88XX family phones release 14.3(1) and above
- Expressway 12.7 (AR/SK)
- Unified CM 14 (SK)
- Other devices support failover in a limited way due to the absence of STUN Keepalives and Adaptive Routing

Endpoints Without Adaptive Routing and STUN Keepalive

- Failover behavior always dictated by the construction of pre-canned routes (through Route Headers)
- Pre-canned routes statically configured by the endpoint, and completely disjoint (for instance, EXPC2 can't be in the pre-canned route EXPE2-EXPC2-CUCM2 and in the EXPE3-EXPC2-CUCM3)



- Inter-cluster failover achievable only if the number of CUCM nodes in the CallManager group $\leq$ number of Expressway-C peers = number of Expressway-E peers

How to Address the “Endpoints Are Not Failing Over” Issue (1)

- Check if Expressway supports AR/SK (12.7 and above)
- Check if UCM supports SK (14 and above)
- Check if the endpoint model/release supports Adaptive Routing and STUN Keepalives
- In case AR and SK are not supported, make sure that the number of Unified CM in the CallManager Group of the phone is equal or smaller than the number of Expressway-C, and the number of Expressway-E

How to Address the “Endpoints Are Not Failing Over” Issue (2)

- If you are unsure if the endpoint supports AR/SK:
 - Turn on Diagnostic Logging on Expressway-E
 - Start the client/endpoint and wait for the successful registration
 - Stop and collect the logs
 - Look for “REGISTER sip:” in the Expressway logs and locate the right record. This tells you everything you need to know about your endpoint failover capabilities
 - Contact Header in REGISTER message shows AR and/or SK support:

```
Contact: <sip:a4d71c16-61af-06b2-3ddb-9b44c50384b6@192.168.1.224:57851;transport=tls>;+sip.instance="<urn:uuid:00000000-0000-0000-0000-0000-f82fa8e4a606>";+u.sip!devicename.ccm.cisco.com="CSFagoodman";+u.sip!model.ccm.cisco.com="503";sip.ice;video;bfcpx-cisco-mra-ha=AR_SK;x-cisco-reg-id=1
```

- `x-cisco-mra-ha={AR, SK, AR_SK}`

Troubleshooting Example (1)

“This endpoint does not fail over”: CUCM 14 and Expressway 14

```
SIPMSG:
|REGISTER sip:ccm002.cloudentpa.com SIP/2.0
Via: SIP/2.0/TLS 203.0.113.241:7001;egress-zone=ZoneToC;branch=z9hG4bKa5d15b9584dee859d4544f4570f1aa16477.f6e495f6ceab4be6b6c300415dec7607;proxy-call-id=2b1d1a67-9f01-4563-899c-79fb91690a11;rport
Via: SIP/2.0/TLS 192.168.220.50:52094;branch=z9hG4bK7aeab4a7;received=198.51.100.150;rport=51413;ingress-zone=CollaborationEdgeZone
Call-ID: 00c1b1fd-74560003-69c02052-4e1a836b@192.168.220.50
CSeq: 336 REGISTER
Contact: <sip:9ddfd672-5215-25af-fd02-b32e07cb4c12@192.168.220.50:52094;transport=tls>;+sip.instance="urn:uuid:00000000-0000-0000-0000-00c1b1fd7456";+u.sip!devicename.ccm.cisco.com="SEP00C1B1FD7456";+u.sip!model.ccm.cisco.com
From: <sip:1012@ccm002.cloudentpa.com>;tag=00c1b1fd745600b66224373b-6fb5384e
To: <sip:1012@ccm002.cloudentpa.com>
Max-Forwards: 15
Route: <sip:ccm002.cloudentpa.com;transport=tcp;lr>
Path: <sip:203.0.113.241:7001;transport=tls;lr>
Path: <sip:198.51.100.150:51413;transport=tls;apparent;ds;lr>
User-Agent: Cisco-CP8865/14.2.1
Expires: 3000
Date: Thu, 23 Oct 2025 14:21:16 GMT
Supported: replaces,join,sdp-anat,norefersub,resource-priority,extended-refer,X-cisco-callinfo,X-cisco-serviceuri,X-cisco-escapecodes,X-cisco-service-control,X-cisco-srtp-fallback,X-cisco-monrec,X-cisco-config,X-cisco-sis-7.0.0
P-Asserted-Identity: <sip:1012@ccm002.cloudentpa.com>
X-TAATag: 560401ce-4091-40ed-a377-40ec72d373bf
Session-ID: c32f20e200105000a00000c1b1fd7456;remote=8a7ebd5d00255000a00000c4e1e00000
Content-Length: 0
```

- x-cisco-mra-ha=AR_SK not in Contact Header
 - User-Agent: 8865 phone ✓
 - 8865 Release: 14.2.1 ✗
 - Minimum release: 14.3.1

Troubleshooting Example (2)

“This endpoint does not fail over”

- The endpoint does not support AR/SK
- CallManager Group for the endpoint: ccm001, ccm02, ccm03
- Expressway cluster made of 1 x Expressway-C and 1 x Expressway-E
- Result: when the MRA Unified CM goes down, the endpoint does not fail over to another Unified CM in its CallManager Group.
- What are the possible solutions?
- Solutions:
 - 1. Add 2 more Expressway-C and Expressway-E (4 servers)

OR

- 2. Add 1 Expressway-C and 1 Expressway-E (2 servers) and remove ccm03 from the CallManager Group

Provisioning and Registration Load Balancing

- Expressway-E load balancing through DNS SRV priority and weight
- SIP Registration, XMPP Registration load-balanced through:
 - CallManager Groups on UCM (static)
 - Expressway-C response to get_edge_config request (dynamic)
- External load balancers may interfere with the Expressway's native load-balancing logic and are not supported

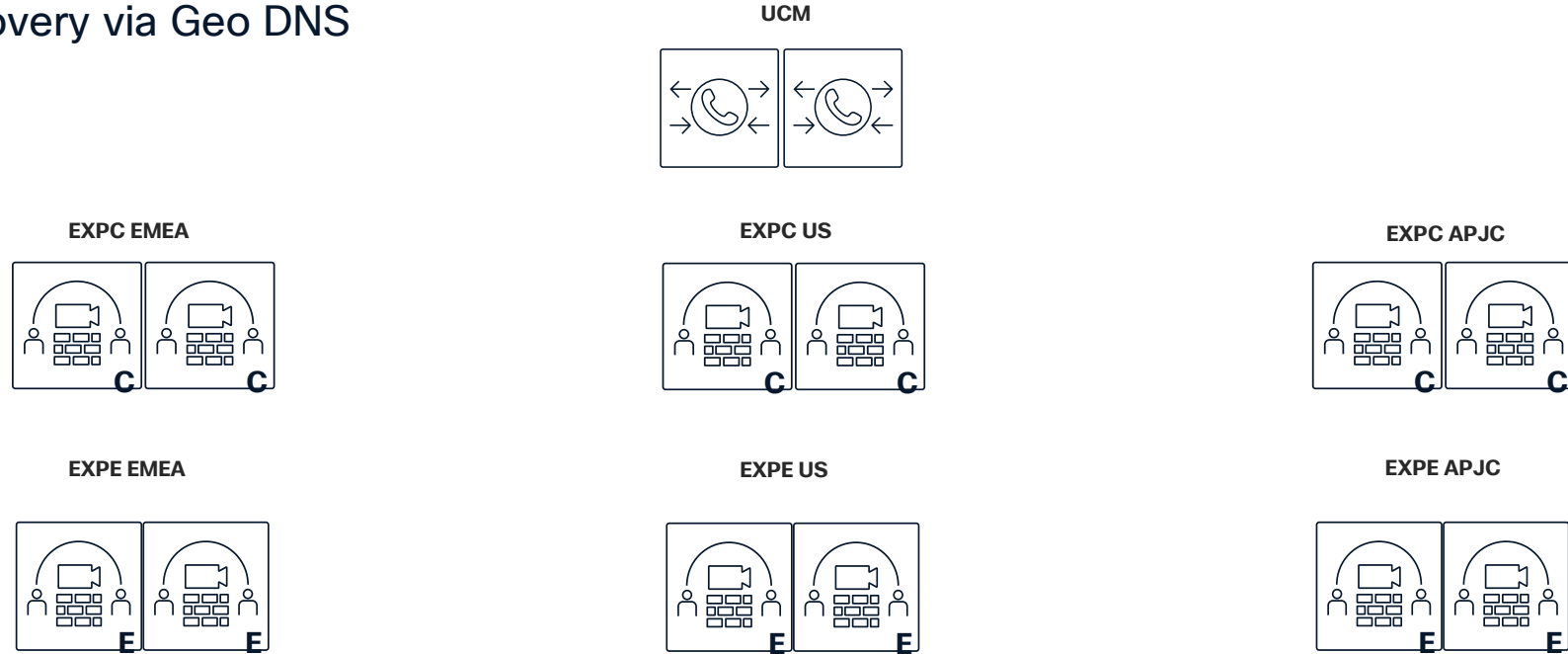
Large Scale Design

Key Points

- Centralized or Distributed Unified CM Clusters
- Regional Expressway Clusters
- Geo DNS For Service Discovery
- Geo DNS For Expressway-E Datacenter Redundancy
- Expressway-C Load Balancing for SIP/XMPP Managed by Expressway-C Through *getEdgeConfigResponse*
- Unified CM Load Balancing Through Unified CM Groups

Distributed Expressway Clusters Per Region

Service Discovery via Geo DNS



- Users in EMEA register through EMEA cluster – backup in US and APJC
- Users in US register through US cluster – backup in APJC and EMEA
- Users in APJC register through APJC cluster – backup in EMEA and US

DNS SRV Example

Example

- EMEA Region – backup in US and APJC

```
_collab-edge._tls.example.com 10 10 8443 emea_expe1.example.com  
_collab-edge._tls.example.com 10 10 8443 emea_expe2.example.com
```

```
_collab-edge._tls.example.com 20 10 8443 us_expe1.example.com  
_collab-edge._tls.example.com 20 10 8443 us_expe2.example.com  
_collab-edge._tls.example.com 20 10 8443 apjc_expe1.example.com  
_collab-edge._tls.example.com 20 10 8443 apjc_expe2.example.com
```

} Normal working behavior

} If no Expressway-E peers are available in EMEA, clients will be distributed across the US and APJC cluster peers.

Datacenter Failover

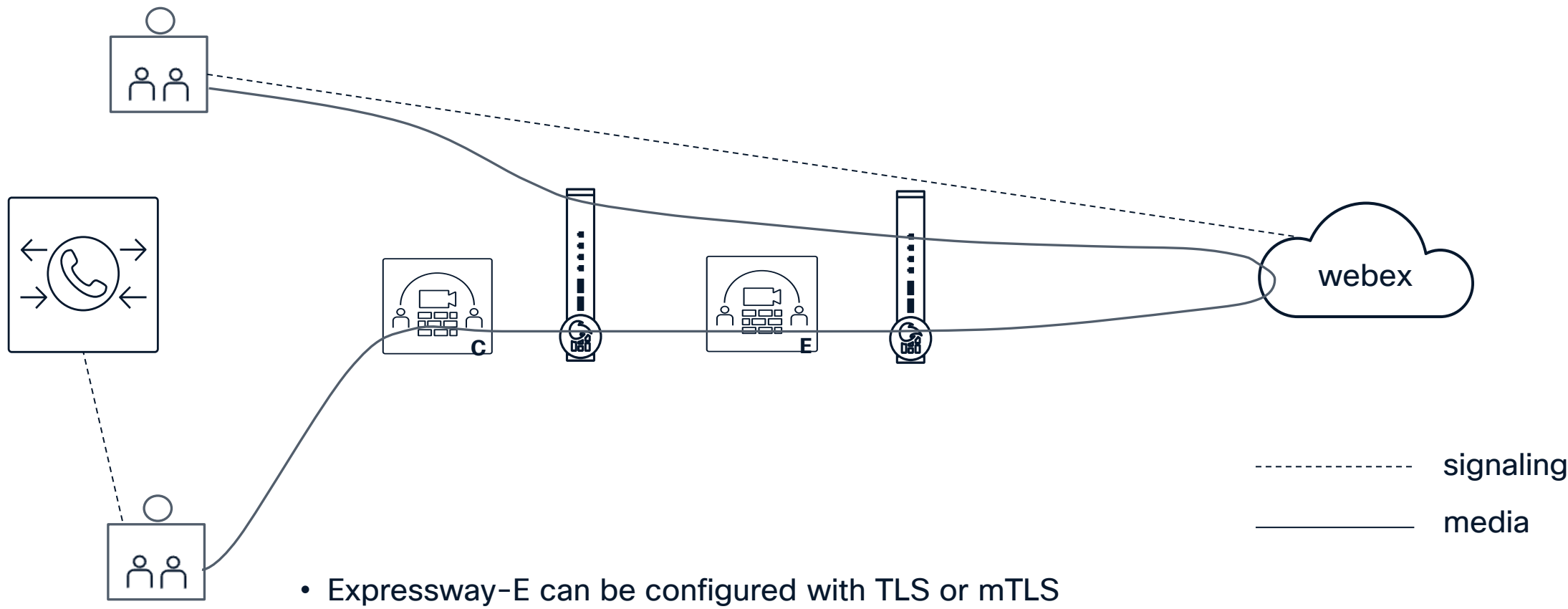
- In case one of the datacenter goes down, the other Expressway clusters act as backup
- Clients are spread across one or more backup Expressway clusters
- Load distribution in backup datacenters is statistical (based on DNS SRV Priority)
- Inter-cluster feature to be released for Jabber and Webex app
- Endpoints might require a reboot
- Rehomeing to the original cluster once the datacenter is restored is not immediate. This happens when the clients perform a new service discovery

Cloud And On-Premises Endpoints

Mixed Calling Scenario

- On-prems Devices calling Cloud devices is still a supported scenario
- SIP TLS
- Expressway certificate signed by a Webex trusted public CA:
<https://help.webex.com/en-us/article/0q4utq/Supported-Certificate-Authorities-for-Webex-Hybrid-Services>
- B2B Architecture Details covered in old BRKCOL-2018 Cisco Live Sessions
- Preferred Architectures, Chapter “Collaboration Edge” for details

UCM Devices and Cloud Devices



Client Authentication ECU Sunset

Digital Certificates and EKU

What is a Digital Certificate?

A digital certificate is an electronic credential issued by a trusted Certificate Authority (CA) that:

- binds a public key to an identity and
- enables authentication and integrity
- In protocols such as TLS, it allows peers to verify identities and securely establish encrypted sessions.
- Commonly used for TLS, code signing, and email security

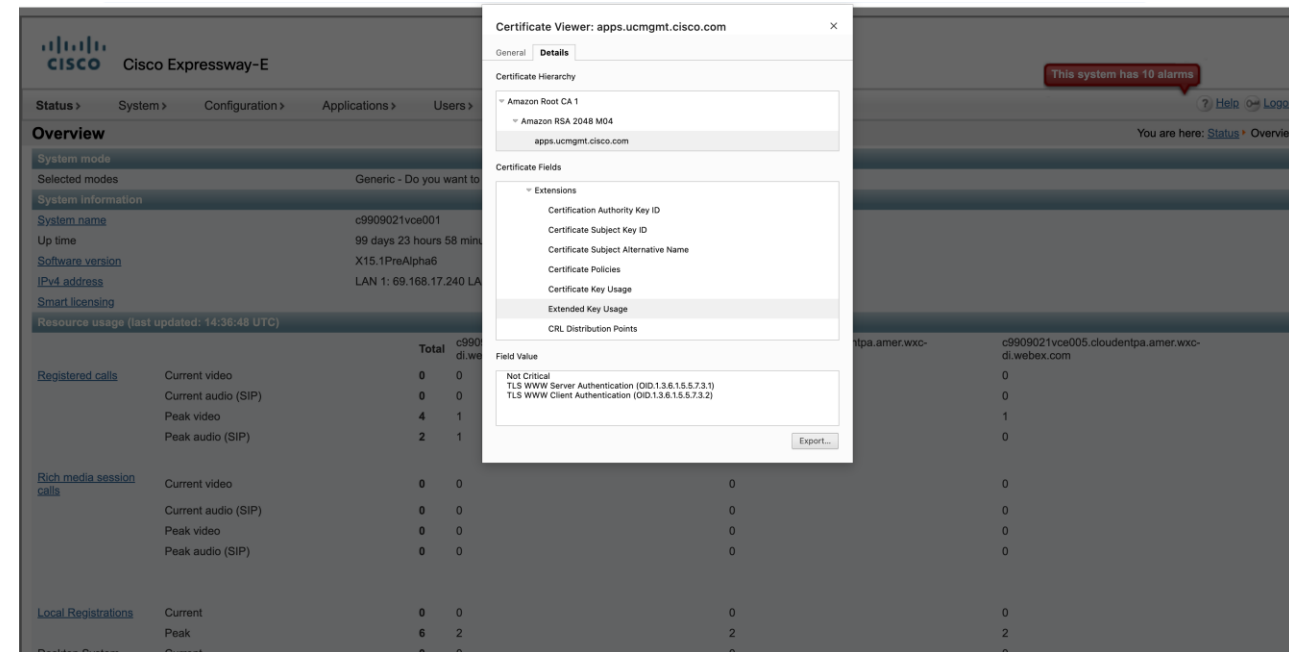
A certificate contains:

- Subject information (the certificate holder)
- A public key
- Issuer information (the CA)
- A validity period
- Extensions such as Extended Key Usage (EKU)

What is EKU (Extended Key Usage)?

EKU is an X.509 certificate extension that specifies intended purposes for the certificate.

- It restricts how the certificate can be used beyond the basic key usage.
- Some examples of EKU values:
 - Server Authentication (id-kp-serverAuth) → For HTTPS servers.
 - **Client Authentication** (id-kp-clientAuth) → **For mutual TLS clients.**
 - Code Signing → For signing software.
 - Email Protection → For securing emails.

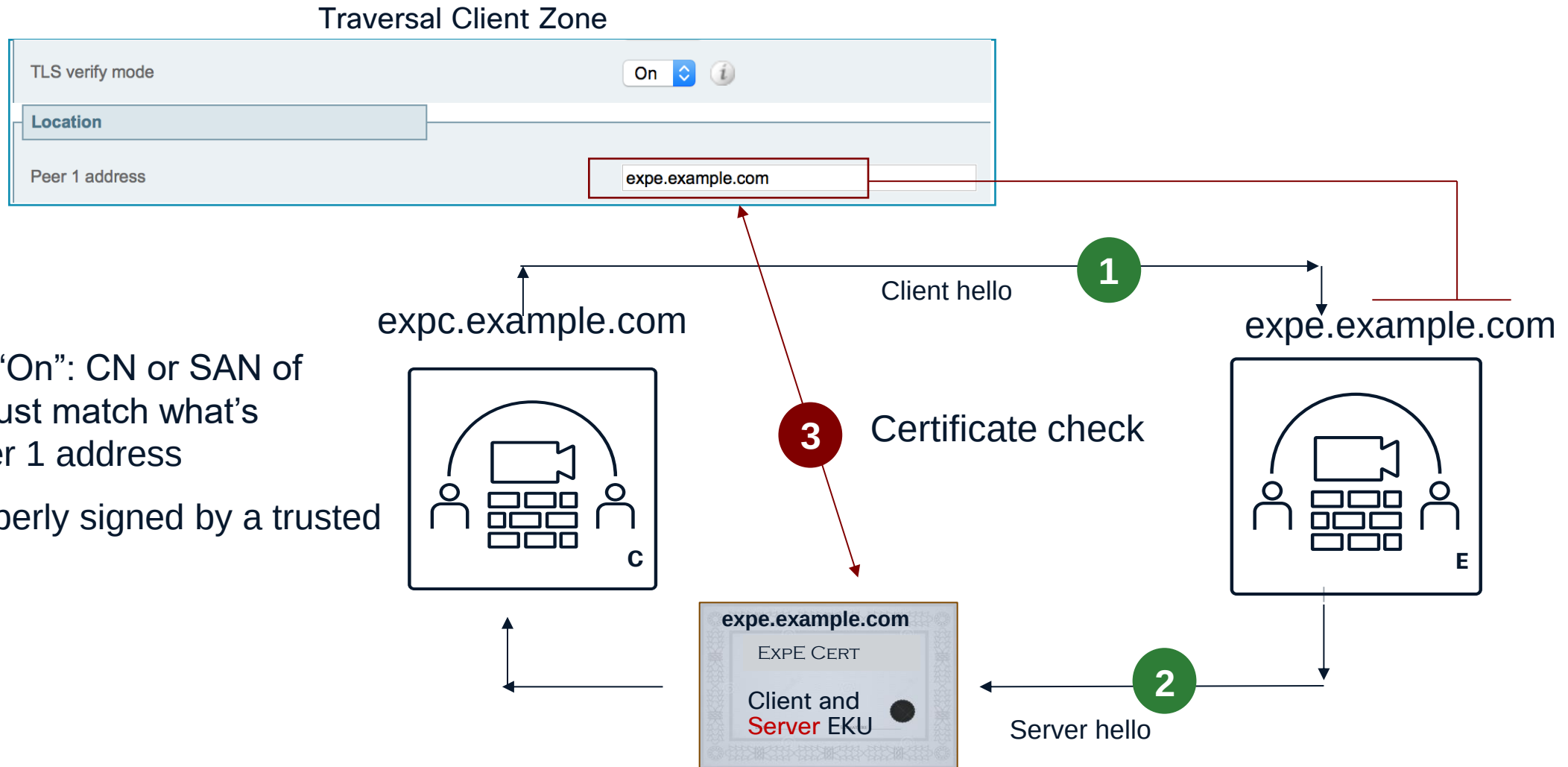


How EKU is Used

Browsers, operating systems, and applications check EKU before trusting a certificate for a specific function.

- If EKU does not match the required purpose, the certificate is rejected for that use.
- Example:
 - A certificate with only clientAuth EKU cannot be used for a web server.
 - A certificate with serverAuth EKU is valid for HTTPS.

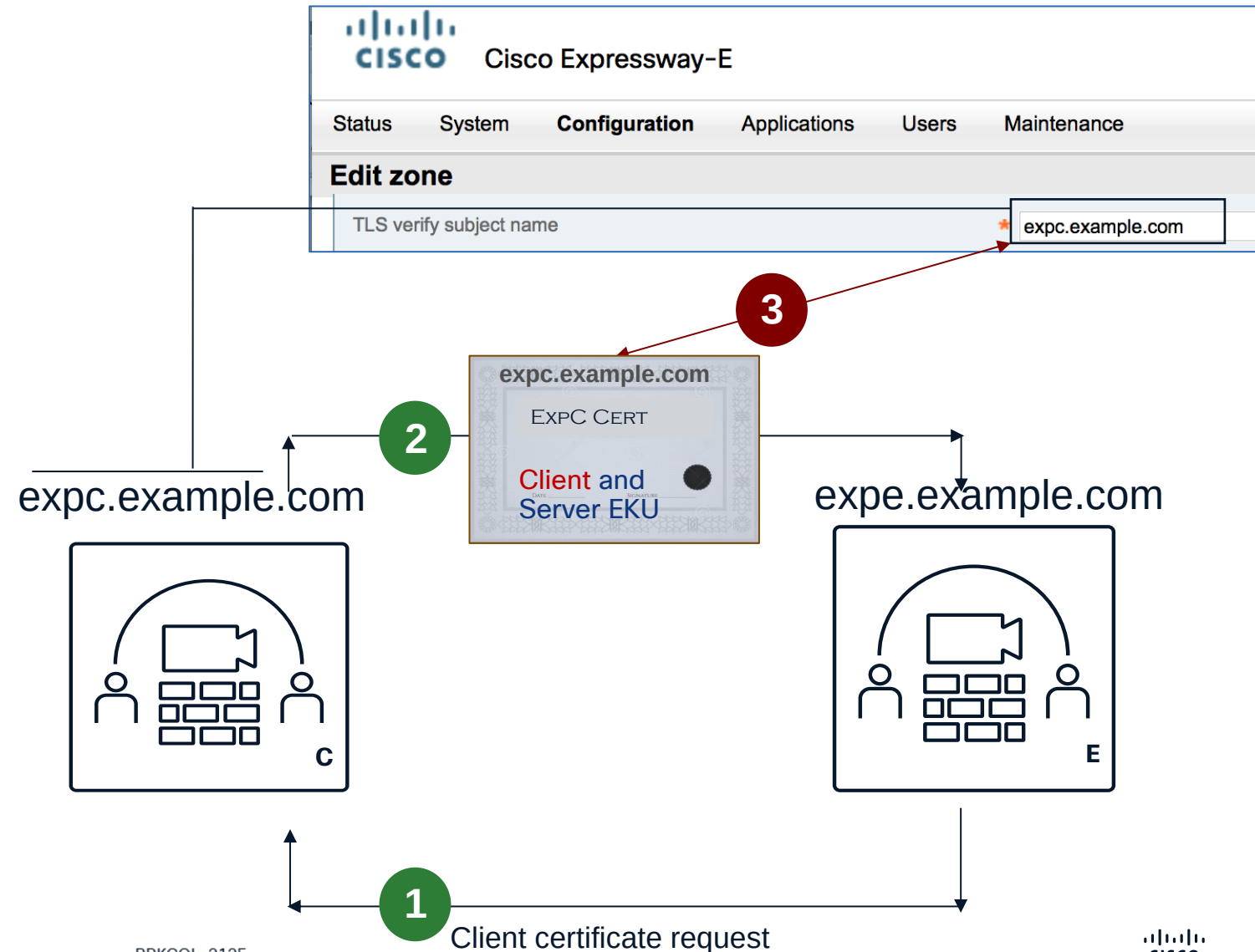
TLS



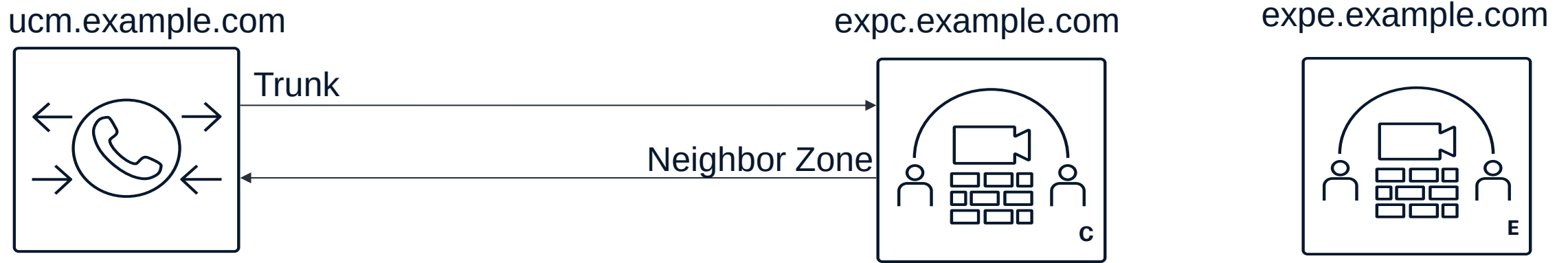
- TLS verify set to “On”: CN or SAN of Expressway-E must match what’s configured in Peer 1 address
- Cert must be properly signed by a trusted CA

mTLS: Client Authentication Phase

- Server auth first (previous slide), then client
- TLS verify set to “On” triggers client certificate request
- CN or SAN of Expressway-C must match what’s configured in “TLS verify subject name”



mTLS in Collab Appliances



- Neighbor Zone and UCM Trunks are created on top of multiple TCP connections
- In the example, one is from UCM to Expressway-C, another from Expressway-C to UCM
- In each connection there is a TLS client and a TLS server
- The server sends the certificate to the client
- The client sends the certificate to the server only if requested (mTLS enabled)

Chrome Root Policy

The Chrome Root Program Policy establishes the minimum requirements that a root Certification Authority (CA) certificate must meet to be included and remain trusted by default in the Google Chrome web browser's Root Store. (Chrome Root Program Policy)

- Core Requirements for Trusted Root CAs
- CAs must comply with CA/Browser Forum Baseline Requirements and Chrome-specific checks.
- No more mixed-use roots; **ServerAuth-only** hierarchies required for public server TLS by **June 15, 2026**
 - **ClientAuth ECU Sunset:** All new public TLS server certificates must **omit clientAuth** by **June 15, 2026**, enforced gradually starting mid-2025
- Chrome Root Policy is followed by Chromium-based Browsers (Eg. Chrome, Edge, Opera, Brave, Samsung)

Date/Period	Event	Policy Detail & Impact
May 2026 (Approx.)	CA Issuance Cut-off	Major CAs will fully remove the option to issue new public TLS certificates containing the Client Authentication ECU. This is the last point CAs will issue dual-purpose certificates.
June 15, 2026	Chrome Distrust Deadline	Chrome will stop trusting any public TLS server certificate (leaf certificate) that contains both the Server Authentication and Client Authentication

Response from Public CA providers

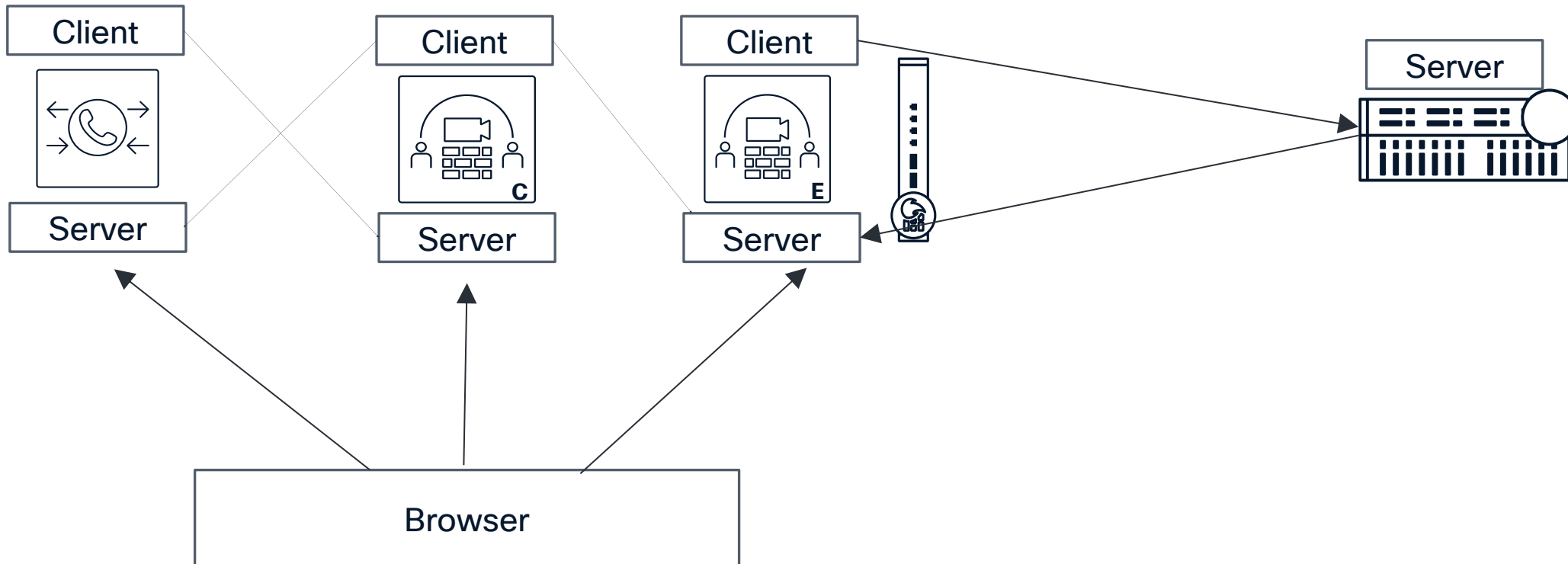
- Sept-Oct 2025
 - Many public CAs began *not including clientAuth EKU by default* for newly issued public TLS certs.
 - Examples: DigiCert, Sectigo, SSL.com, they shifted to serverAuth EKU only by default and require explicit request to include clientAuth during an interim period.
 - Typical vendor notes: Oct 1, 2025 (DigiCert start default change); Sept/Oct 2025 timelines
 - from others.
- May 1 – Jun 15, 2026
 - Public CA vendors plan to remove the option to issue new public TLS certs with clientAuth EKU entirely
 - Dates may vary: some vendors start by May 1 / May 15, others point to industry enforcement around Jun 15, 2026
 - Certificates issued before these cutoff dates generally remain valid until they expire.
 - After these dates, public CAs will no longer issue TLS certs usable for client authentication
 - Organizations must migrate to private CAs or alternative authentication methods for mTLS/client-auth use cases.

Chrome Root Policy

- Any CA that is not compliant to this policy won't be added in Chrome Root Store
- What this does not mean:
- If a CA releases a certificate with EKU set to Client and Server, it won't be included by default in Chrome Root Store, but it can be added to it
- Private CA can still sign certificates with both EKU Client and Server validation, provided that its root CA is added to the Chrome Root Store

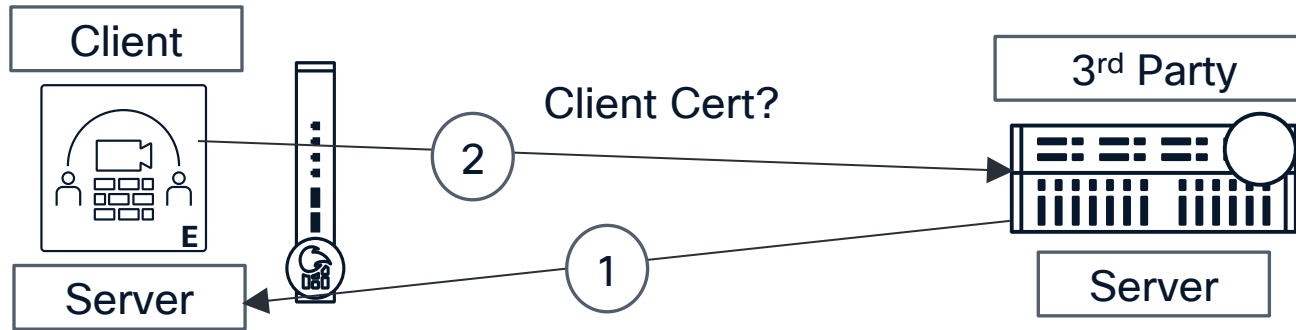
Dual Certificate Feature Enhancement (X15.5)

- Expressway (15.5) will allow 2 certificates, a client ECU certificate, and a server ECU certificate
- Based on directionality of the TLS connection, Expressway will use one or the other
- The following picture shows the TLS connections with dual certificates in each box



Ignoring Client EKU Validation

- Use Cases: mTLS where the remote party has a certificate with Server only option



- If a certificate with an EKU that does not correspond to the TLS role is received, then the TLS connection will be aborted
- It is possible to turn off client EKU validation and have this scenario work
- It won't work if the certificate has Server validation and it is used as Client

Let's Encrypt

- Let's Encrypt delivers a server-only certificate from February 11th
- Disable automatic renewal of certificates if your Expressway release is not X15.4
- Check the expiration time of the current certificates

Expressway Release X15.4

- Expressway release X15.4 allows to use the HTTP interface to import a certificate without ClientAuth EKU
- Impact: by replacing Expressway-E server and client EKU certificate with another one that has ServerAuth EKU only, the following scenarios might not work:
 - Expressway-E calling a remote entity that requests client certificate (mTLS)

Expressway X15.5

- X15.5 will be available in May 2026
- Will allow to host 2 certificates, one for the server and another for the client ECU.
- If X15.4 partial fix doesn't work for your environment, renew the certificate before the CA stops releasing client and server ECU

Security

Use Dedicated Management Interface

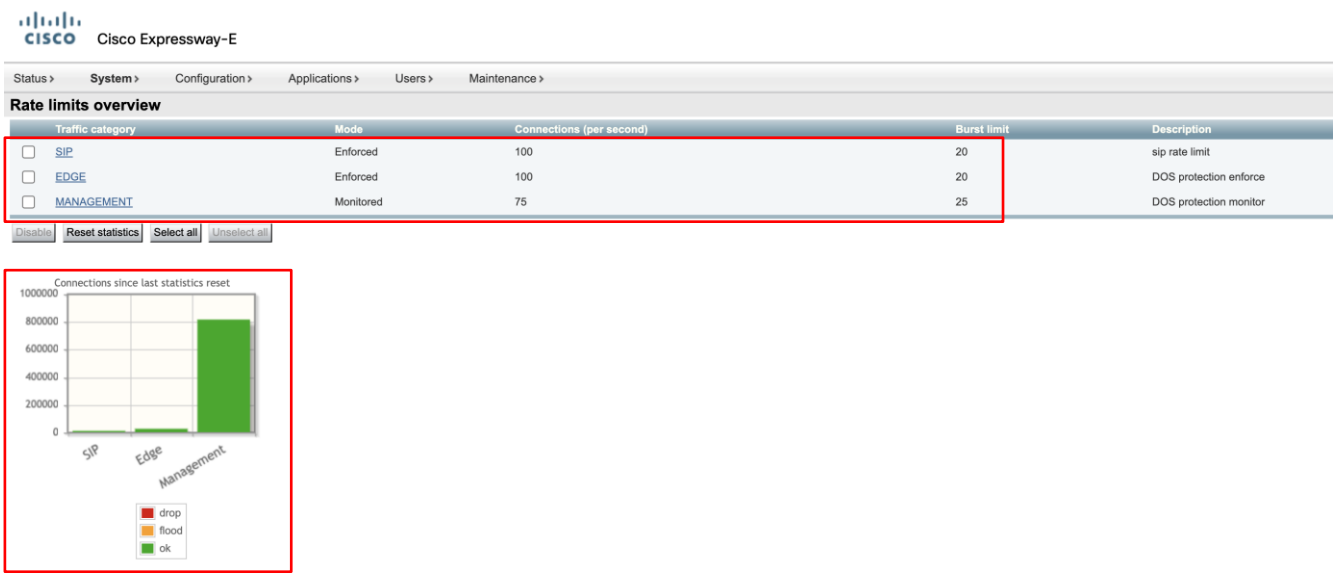
- Make DMI Sole Interface for Management Traffic
 - Management traffic sent and received through LAN3 and no other traffic uses that interface
 - Management traffic allowed only from internal network on DMZ Firewall
 - Management traffic blocked in the firewall if coming from LAN1 (internal network) or LAN2 (external interface)
 - Use the Expressway firewall if policies can't be applied to the Internet/DMZ firewall

Host-Based Intrusion Protection

- Expressway includes a host-based intrusion protection system (fail2ban)
- When enabled, Expressway scans logs for signatures, and establishes dynamic rules to block source ip addresses
- This protects against brute force attacks that originate from a single source ip address
- When a blocked IP tries to access the system, the request is dropped by the firewall and no response is sent back to the source

DoS Protection for Expressway

SIP - tcp: 5060,5061,5062, udp: 5060
Edge - tcp: 8443
Management - tcp: 22,80, udp: 161, tcp: 443



Automated Detection on Expressway-E

Intrusion Prevention automatically bans IP's matching the rules in each category

Automated detection overview

You are here: [System](#) > [Protection](#) > [Automated detection](#) > [Configuration](#)

Category	Status	Currently blocked	Total failures	Total blocks	Exemptions	Action
☐ External API authorization failure	Off	-	-	-	0	View/Edit View exemptions
☐ HTTP proxy authorization failure	✓ On - Active	0	0	0	0	View/Edit View exemptions
☐ HTTP proxy protocol violation	✓ On - Active	0	0	0	0	View/Edit View exemptions
☐ HTTP proxy resource access failure	Off	-	-	-	0	View/Edit View exemptions
☐ SIP authentication failure	✓ On - Active	-	-	-	0	View/Edit View exemptions
☐ SIP protocol violation	Off	-	-	-	0	View/Edit View exemptions
☐ SIP registration failure	✓ On - Active	0	0	0	0	View/Edit View exemptions
☐ SSH authorization failure	✓ On - Active	0	0	0	0	View/Edit View exemptions
☐ SSH protocol violation	✓ On - Active	0	0	0	0	View/Edit View exemptions
☐ SSH tunnels authorization failure	✓ On - Active	0	2	0	0	View/Edit View exemptions
☐ SSH tunnels protocol violation	Off	-	-	-	0	View/Edit View exemptions
☐ Web interface authorization failure	Off	-	-	-	0	View/Edit View exemptions
☐ Web interface protocol violation	Off	-	-	-	0	View/Edit View exemptions
☐ XMPP protocol violation	✓ On - Active	0	0	0	0	View/Edit View exemptions

[Enable](#) [Disable](#) [Reset counters](#) [Select all](#) [Unselect all](#)

All statistics are for this peer only

MRA User Authentication – HTTP 401, HTTP 407 Errors

Spam Calls Mitigation – 403 Forbidden

Registration Rejected

'Monitors invalid SIP messages'

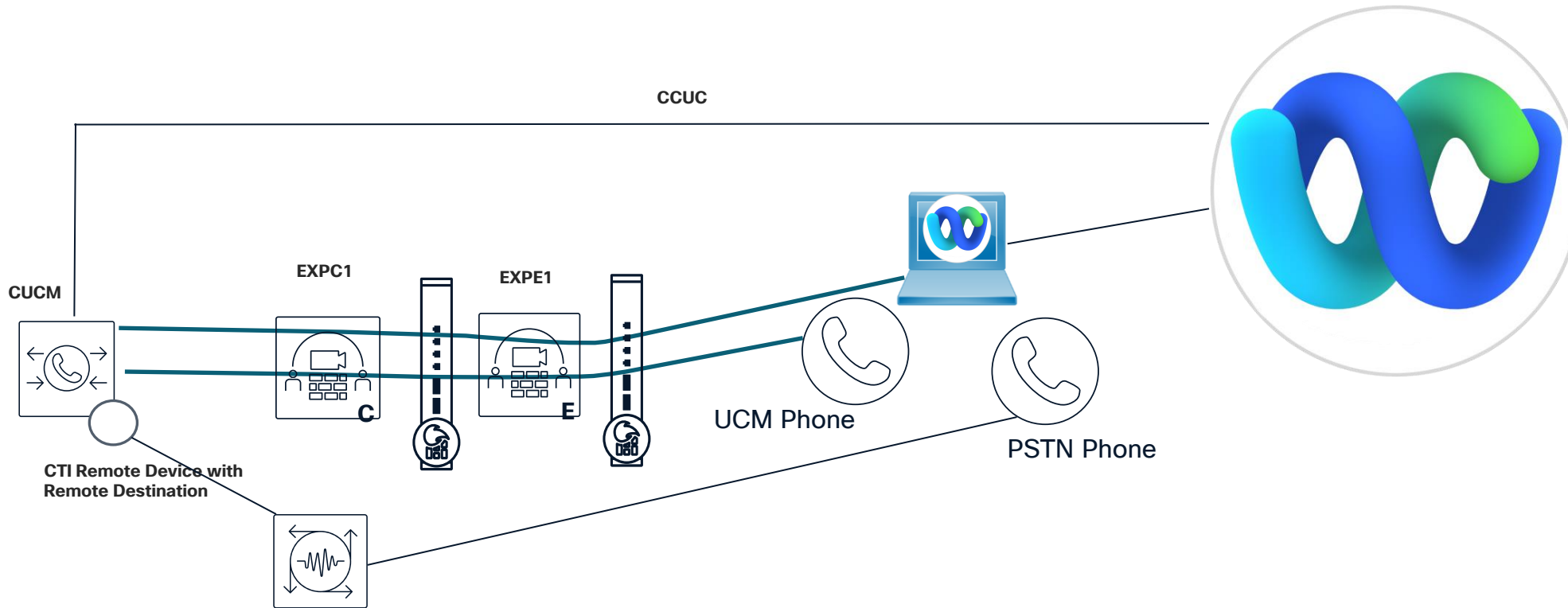
More Security Guidelines

An Expressway Security Presentation can be found here:

<https://www.ciscolive.com/c/dam/r/ciscolive/global-event/docs/2023/pdf/BRKCOL-3301.pdf>

Mobile and Remote Access With CTI

Borderless CTI for Remote Desktop Control



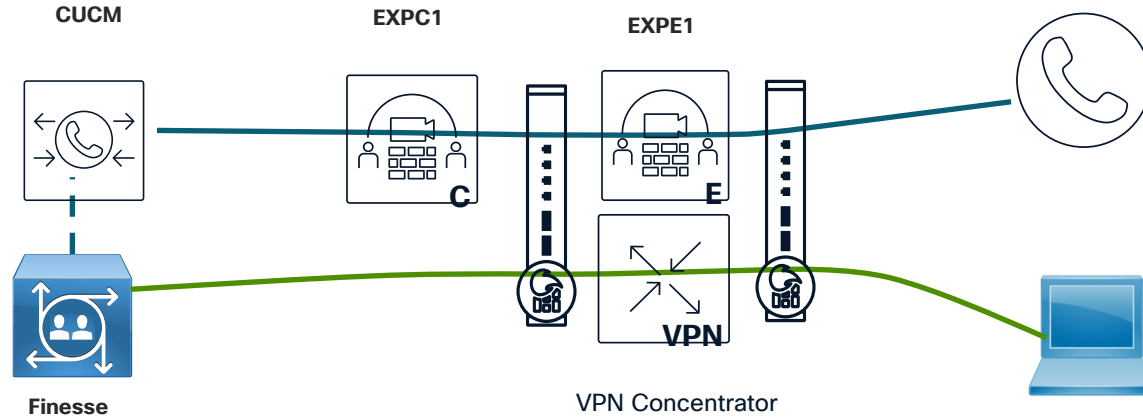
- CUCM Enrolled to the Cloud (CCUC)
- Webex app registered on UCM
- Webex app capable of CTI-controlling the associated device through CUCC
- Extend and Connect for PSTN phones

CTI over MRA

- General-purpose CTI is not supported over MRA
- For this reason, applications using CTI need different mechanisms
 - VPN
 - Finesse VPN-less

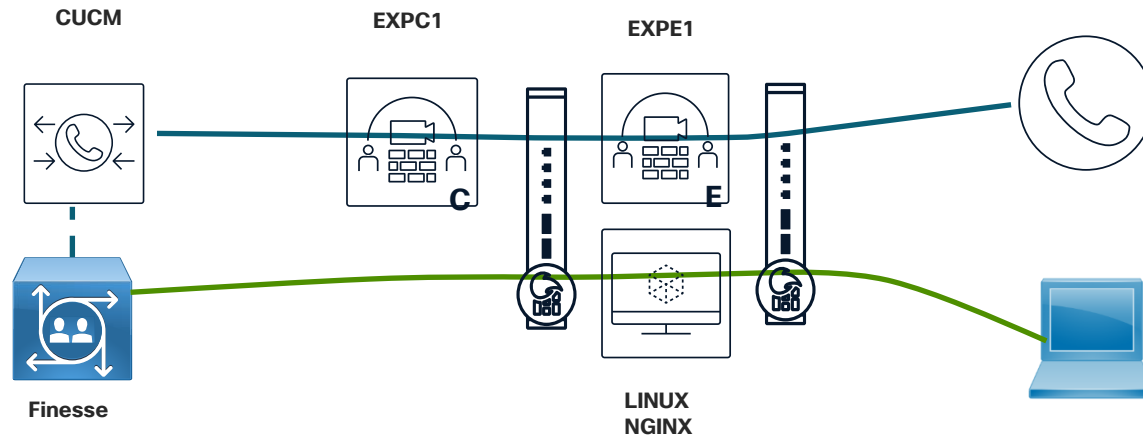
UCCX With Expressway and VPN Concentrator

- Finesse desktop on VPN, phones via MRA



UCCX With Expressway and HTTPS Reverse Proxy

- Finesse desktop via HTTPS Reverse Proxy, phones via MRA



VDI Architectures

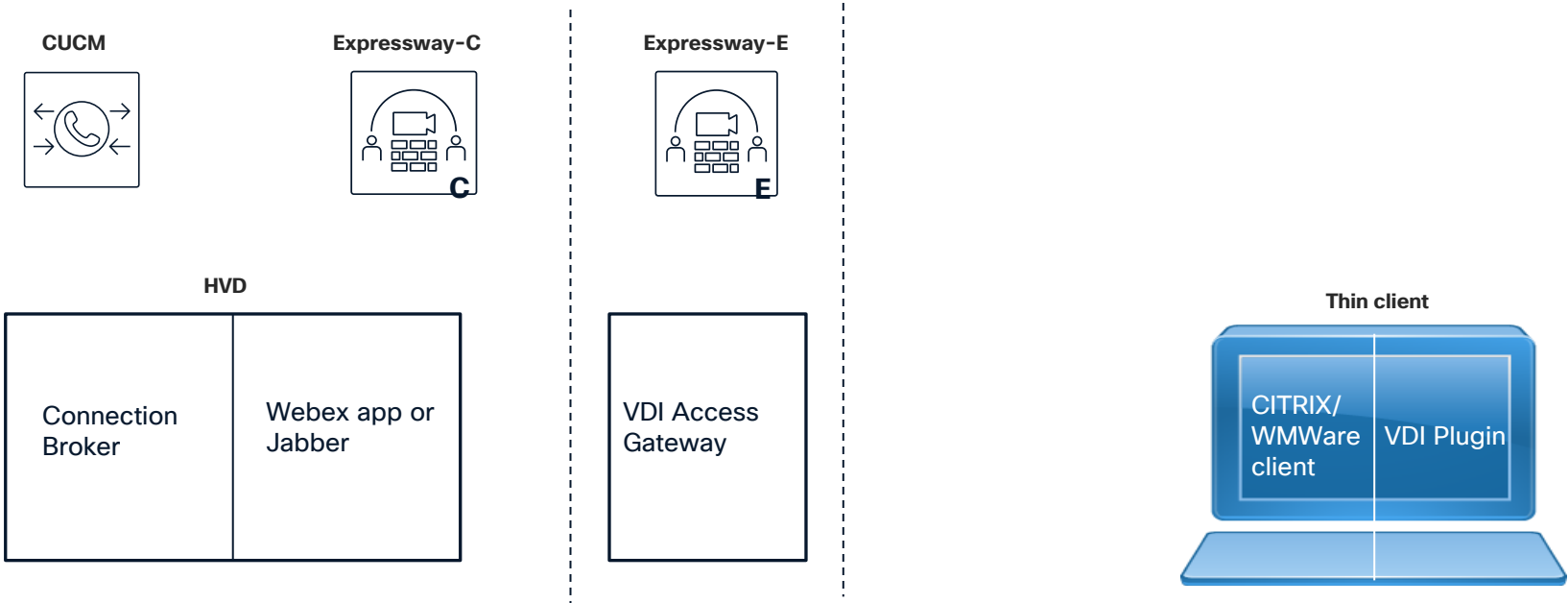
VDI Architectures

- The objective of VDI architectures is to enable the use of thin clients, with all application processing performed on Hosted Virtual Desktops (HVDs) deployed in the datacenter.
- While messaging functions operate well within the standard VDI architecture, media traffic (voice and video) can cause a "hairpin effect," where media streams loop through the datacenter, potentially saturating bandwidth.
- To mitigate this, Cisco has developed optimized VDI architectures for Jabber and Webex app that offload media processing to the local thin client, enabling direct media paths between endpoints and avoiding the hairpin effect. This approach preserves bandwidth and improves media quality.

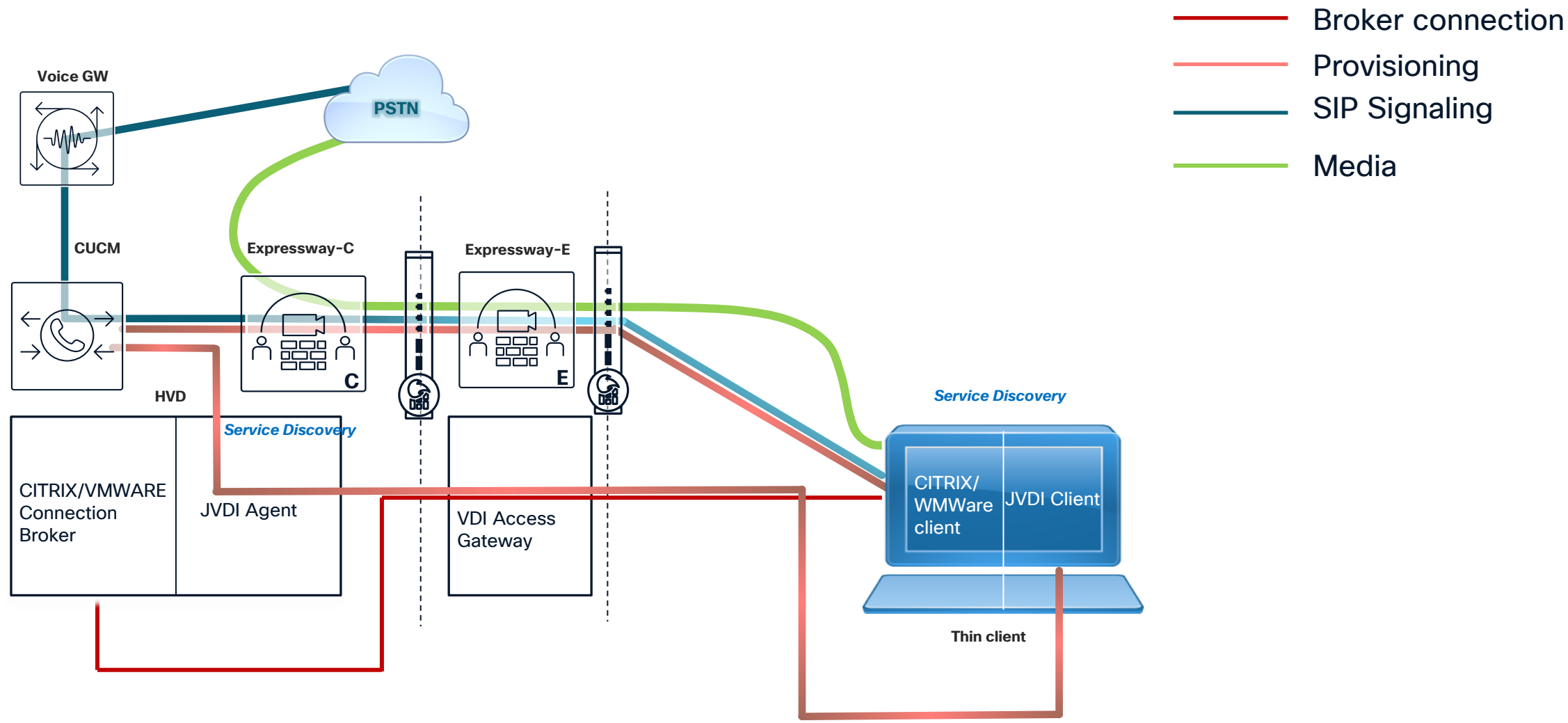
VDI Environment Overview

- **Connection broker** - The resource and connection manager (Citrix, VMware) that connects the virtualization provider to the thin client. Manages user authentication and connection to virtual desktops
- **Hosted virtual desktop (HVD)** - The Virtual Machine (VM) that runs the Webex App/Jabber remotely.
- **Thin client** - The endpoint (for example, low profile or repurposed desktop or laptop) that the user uses to sign in remotely to the virtual desktop. The Citrix Workspace App or VMware Horizon Client and the VDI plugin are installed on this device to access the HVD environment.
- **VDI Plugin** - Software that is installed on the thin client used to offload media processing to the user device instead of HVD resources (**Citrix Workspace App** and **VMware Horizon Client**)
- **Access Gateways** - If the thin client is off-premises, then Citrix/VMware access gateways are required to connect to the HVD

VDI Architectural Diagram



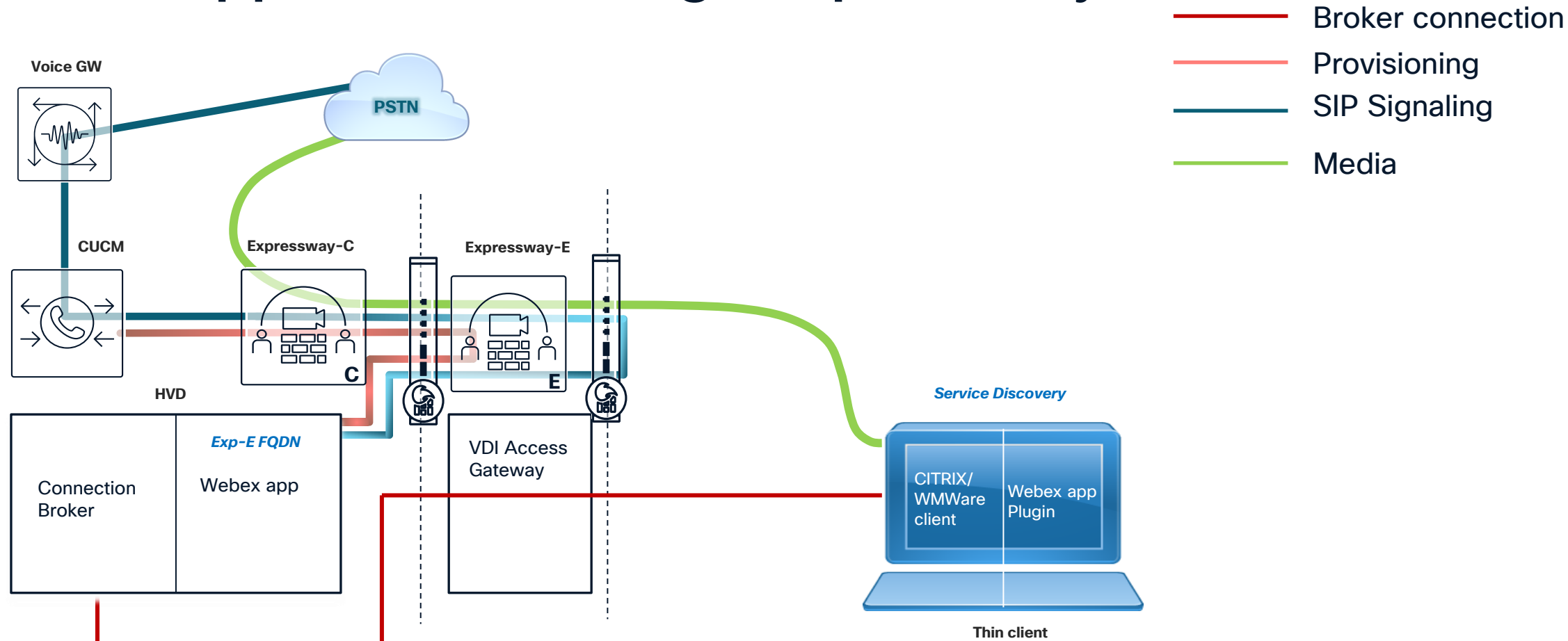
Jabber for VDI Through Expressway



Webex App for VDI

- The main objectives of Webex App for VDI are:
 - Offloading media traffic from central servers
 - Keeping the VDI plugin on the thin client lightweight
- To achieve the above tasks, provisioning and signaling are performed by the HVD, while media is directly managed by the VDI Plugin

Webex App for VDI Through Expressway



Expressway-E FQDN DNS Resolution

Internal Network	Internet
Expressway-E internal private address	Expressway-E external public address

Webex App for VDI

Summary

- Expressway-E FQDN resolves with the non-natted IP address of the internal interface in the internal network
- Expressway-E FQDN resolves externally with the natted IP address of the external interface
- Provisioning and signaling are managed by the HVD
 - Provisioning through the internal interface of Expressway-E. This interface must be reachable by the HVD
 - SIP Signaling through the external, natted interface of Expressway-E
 - Media from the thin client and the Expressway-E (Optimization)

Resources

- Admin Guide
https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/expressway/admin_guide/X15-4/exwy_b_cisco-expressway-administrator-guide-x154.pdf
- Virtualization
https://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/uc_system/Solution-guide-virtualization-guide/Cisco-Virtualization-Guide-for-Cisco-On-premises-Calling-Applications.pdf
- Architecture
<https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Collaboration/enterprise/14/collbcvd.html>
- ECU ClientAuth Sunset
<https://www.cisco.com/c/en/us/support/docs/field-notices/743/fn74362.html>
<https://blog.webex.com/collaboration/certificate-ecu-changes-actions-cisco-premises-collaboration-customers-must-take-now/>
<https://help.webex.com/en-us/article/zo55dl/Public-CA-certificate-changes-impacting-Dedicated-Instance>
- Borderless CTI
<https://help.webex.com/en-us/article/00tesu/Enable-Borderless-CTI-through-Webex-Cloud-Connected-UC>

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